

APEX: AI Platform of Platforms

Cloud-Native AI Agent Architecture | TOGAF 10 + AI-DLC

A comprehensive TOGAF 10 Architecture Development Method (ADM) guide for designing, governing, and delivering an AI Platform of Platforms using cloud-native, vendor-agnostic open standards.

Integrates the AWS AI-Driven Development Lifecycle (AI-DLC) methodology with full impact analysis across all nine ADM phases and operating model adaptations required for the AI-native enterprise.

ORGANISATION

GlobalCorp Enterprises

FRAMEWORK

TOGAF 10 ADM + AI-DLC

DOCUMENT REF

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CLASSIFICATION

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APEX — AI Platform of Platforms

CLOUD APPROACH

Cloud-Native / Vendor-Agnostic

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APPROVED

OVERVIEW

Executive Summary

Context, scope, and key programme facts

About This Document

This document is the authoritative Enterprise Architecture (EA) reference for the APEX programme at GlobalCorp Enterprises. It applies the **TOGAF 10 Architecture Development Method (ADM)** across all phases to architect, govern, and deliver a unified AI Platform of Platforms using cloud-native, vendor-agnostic open standards. It integrates the **AI-Driven Development Lifecycle (AI-DLC)** — introduced by AWS in July 2025 — analysing its impact on every ADM phase, enterprise operating model, and governance structure.

All technology choices are documented using **cloud-native, vendor-agnostic patterns**. Specific managed services may be selected from any hyperscale provider (or a combination) during Phase E/F without requiring architectural rework — the abstractions are intentional.



Programme Pain Chain — Current vs Target

Problem Area	Current State	APEX Target	AI-DLC Acceleration
Time-to-market for AI	9.2 months	2.5 months	AI-DLC bolts compress to <1 month
Duplicated AI tool spend	\$14 M / year	\$3 M / year	Platform reuse + AI-DLC pattern library
Governance gaps (regulatory)	47 open gaps	0 gaps	AI-generated compliance artefacts
Developer satisfaction (NPS)	−12	+55	Mob Construction, self-service portal
Agents in production	23 (siloed)	120 (platform)	AI-DLC Operations phase automation
MTTR for AI incidents	6.3 hours	<30 minutes	AI-DLC real-time monitoring & IaC

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AI-DLC: Research Findings & EA Impacts

AWS AI-Driven Development Lifecycle — implications for TOGAF 10 practice

What is AI-DLC?

The **AI-Driven Development Lifecycle (AI-DLC)** was published by AWS in July 2025 and featured as a headline methodology at AWS re:Invent 2025 (session DVT214). It emerged from over 100 enterprise experiments and observed that both existing AI-in-software approaches — 'AI-assisted' (narrow tasks) and 'AI-autonomous' (full generation) — produced suboptimal velocity and quality. AI-DLC positions AI as a **central collaborator and teammate** throughout every development activity, not a peripheral helper.

Customer results: Wipro completed months of planned enterprise healthcare work in **20 hours** using AI-DLC. Productivity gains of **10–15×** were reported consistently. The Atlassian RovoDev 2026 study found 38.7% of AI code-review comments led to additional fixes; the Qodo 2025 AI Code Quality Report showed quality improvements rising from 55% to 81% with AI-assisted review.

The Three Phases of AI-DLC

Phase	Name	AI Role	Human Role	EA Impact
1	INCEPTION	Transforms business intent into requirements and architecture units via Mob Elaboration	Validates AI questions; provides business context and domain knowledge	Vision, requirements, and capability maps generated in hours not weeks
2	CONSTRUCTION	Proposes architecture, domain models, code, and tests via Mob Construction	Clarifies technical decisions and architectural choices in real time	Phases B/C/D artefacts produced concurrently; ADM phase boundaries blur
3	OPERATIONS	Applies accumulated context to manage IaC and deployments with oversight	Reviews and approves infrastructure changes; monitors production outcomes	Phases F/G/H collapse into continuous delivery; compliance must be automated

AI-DLC vs Traditional Agile — Terminology Changes

Traditional Term	AI-DLC Term	Duration Change	Architecture Implication
Sprint	Bolt	2–4 weeks → hours to days	ADM phases must operate within bolt cadence
Epic	Unit of Work	Months → days to weeks	Architecture work packages shrink; more frequent ARB
User Story	AI-Elaborated Requirement	Days → minutes	Requirements traceability automated
Code Review (PR)	AI-Augmented Quality Scan	Hours → minutes	Automated conformance checking replaces manual PR review
Architecture Review	Real-time Mob Construction	Weeks → concurrent	ARB provides real-time guidance during Construction
Documentation	Persistent Semantic Context	Post-sprint → continuous	Architecture repository auto-updated as artefacts are created
Retrospective	Continuous Learning Loop	End of sprint → continuous	Architecture debt auto-detected and surfaced

AI-DLC Impact Matrix: All 10 TOGAF ADM Phases

The matrix below maps AI-DLC's structural implications to each TOGAF ADM phase, identifying what changes, what risks emerge, and what GlobalCorp must do:

TOGAF Phase	AI-DLC Impact	Key Risk	GlobalCorp Adaptation
Preliminary	AI generates architecture principle drafts; governance rules codified as steering files fed to AI coding agents	Principle quality depends on prompt quality and organisational context	Architects validate all AI-generated principles; formal ratification gate mandatory before use
Phase A — Vision	AI transforms business intent into draft Architecture Vision and Statement of Architecture Work in hours	Stakeholder nuance and political context may be missed by AI	Mob Elaboration session with all stakeholders; AI as scribe, humans as authors
Phase B — Business	AI auto-generates business capability maps from existing documentation; value streams modelled in minutes	Existing documents may be stale or inconsistent; garbage-in, garbage-out risk	Data quality programme is a prerequisite; Semantic Context Building for brownfield systems
Phase C — Data	AI creates logical data models, entity diagrams, and API schemas; prompt catalogs and embeddings become first-class data entities	Model drift introduces new data governance risk not covered by traditional TOGAF data models	Embeddings versioned in Data Mesh; prompt catalog treated as governed data and code artefact
Phase C — Application	AI proposes application component models and cloud-native integration patterns during Mob Construction	AI may propose patterns inconsistent with GlobalCorp architectural standards	Organisational standards loaded into AI context as steering files; ARB reviews all AI-proposed patterns
Phase D — Technology	AI generates Infrastructure-as-Code, security policies, and deployment configurations from architecture context	AI-generated IaC errors may not be obvious without careful review; hallucinated resources	Every engineer must understand every line of AI-generated IaC — AI-DLC core principle; mandatory peer review
Phase E — Opportunities	AI analyses gap reports and proposes solution options and build/buy/reuse recommendations at machine speed	AI may miss regulatory obligations, vendor relationship context, or political constraints	Legal, procurement, and compliance participate in Mob Elaboration for all build/buy decisions
Phase F — Migration	Bolts replace traditional work packages; migration plans generated and updated in near real-time	Over-acceleration may skip mandatory governance gates or compliance checkpoints	Compliance gates remain human-controlled; AI-DLC speed is contained within phase boundaries
Phase G — Governance	AI automates conformance checking against architecture contracts; real-time drift detection replaces periodic reviews	False positive conformance flags may create overhead and alert fatigue	Automated checks inform; humans decide. ARB retains final sign-off authority in all cases
Phase H — Change	AI monitors technology landscape (new model releases, regulatory changes) and flags architecture impacts automatically	Continuous change proposals may create architecture instability if not filtered	Change request filtering: AI flags candidates, humans prioritise; quarterly health check retained
Req. Management	AI generates and maintains requirements traceability from business intent through to deployed code	Requirement drift if AI context window is not carefully managed across sessions	Persistent context stored in version-controlled repository; ARS auto-updated with human review gate

Five Operating Model Transformations Required

1 — Governance Tempo: Quarterly → Near Real-Time

Traditional EA governance operates on quarterly review cycles. AI-DLC's bolt cadence requires architecture decisions within hours. GlobalCorp must implement tiered governance: automated checks run continuously, architect-in-the-loop

reviews occur daily, full ARB convenes weekly not bi-weekly. Research from Staun & Stender (2025) confirms these are 'near real-time decisions, not quarterly review topics.'

2 — EA Artefacts: Static Documents → Living Digital Twin

Forrester (2025) found that AI agents in EA suites are now automating data validation, capability mapping, and artefact creation. GlobalCorp's architecture repository must shift from periodic-update model to a continuously auto-updated environment. The EA team transitions from artefact authors to artefact reviewers.

3 — New Roles: Enterprise AI Architect & Chief AI Ethics Officer

BCG research (2024) found 89% of C-level executives ranked AI/GenAI in their top three strategic priorities. In response, organisations are creating dedicated Enterprise AI Architect roles. GlobalCorp must formalise both the Enterprise AI Architect and Chief AI Ethics Officer positions.

4 — Data Architecture: AI-Native Entities as First-Class Artefacts

TOGAF's traditional data architecture covers conceptual, logical, and physical models. AI-DLC introduces new governed entities: embeddings (versioned vector representations), prompt catalogs (regulated as code + data), feedback logs (agent decision outcomes), and model lineage records. These must be treated identically to traditional data assets.

5 — Business Architecture: Dynamic Capabilities Replace Static Models

Traditional Business Architecture documents stable capabilities. AI-DLC creates capabilities that 'learn and adapt in motion.' GlobalCorp's Business Capability Map must flag 'adaptive AI capabilities' separately, with distinct governance mechanisms for capabilities that evolve continuously.

GlobalCorp AI-DLC Adoption Roadmap

Horizon	Period	AI-DLC Scope	EA Impact	Governance Gate
H0: Observe	Q1 2025	AI-DLC training for architecture team; pilot on 2 work packages	Minimal — existing ADM process retained	CTO sign-off on pilot scope
H1: Build Adoption	Q2–Q3 2025	AI-DLC Construction phase for Pioneer Domains 1 and 2 delivery	Phases C/D artefacts AI-assisted; peer review mandatory for all	ARB approves AI-assisted artefacts
H2: Full Inception	Q4 2025	Mob Elaboration for Pioneer 3–5 requirements and architecture vision	Phases A/B artefacts AI-generated; ARB review process updated	Steering Committee endorsement
H3: End-to-End	Q1–Q2 2026	Full AI-DLC across all APEX workstreams; real-time governance active	Full operating model transformation; EA repository AI-integrated	Architecture Board annual review

PRELIMINARY

Preliminary Phase — Foundation

Architecture capability, principles, and governance structures

The Preliminary Phase establishes GlobalCorp's architecture capability for APEX. Before any platform design begins, governance structures are put in place, the ADM is tailored for regulatory context, and architecture principles that govern every subsequent decision are ratified by the board.

Key Activities

- Define Architecture Capability & Maturity Target (CMMI Level 3 target within 18 months)
- Establish the Architecture Review Board (ARB) and full RACI model
- Tailor the TOGAF ADM for regulatory context: EU AI Act, DORA, GDPR, and sector-specific rules
- Populate the Architecture Repository using open EA tooling (ArchiMate 3.2 + C4 notation)
- Define and ratify Architecture Principles across Business, Data, and Technology domains
- AI-DLC Impact: Codify all standards as AI steering files to enforce consistency in Mob Construction

Architecture Principles — APD-001

DOC REF: APD-001 | v2.1 | APPROVED

Business Principles

ID	Principle Name	Statement	Key Implication
BP-01	AI as Business Capability	Every APEX agent must trace to a measurable business outcome in the Corporate Strategy Map	Every domain must submit a Business Value Case before APEX onboarding is approved
BP-02	Human-in-the-Loop by Default	All AI agents acting on sensitive data or executing high-stakes transactions require a defined human escalation path	Human review workflow is mandatory for all agents above a defined risk threshold
BP-03	Platform Thinking over Point Solutions	No business unit shall procure a standalone AI tool if APEX can provide equivalent capability within 90 days	ARB approval required for any AI procurement outside the APEX platform

Data Principles

ID	Principle Name	Statement	Key Implication
DP-01	Data Sovereignty	Personal and sensitive data processed by AI agents must remain within the contractually agreed jurisdiction	Multi-region deployment required; AI inference endpoints provisioned per jurisdiction
DP-02	Data as Shared Asset	All data consumed by APEX agents is catalogued in the GlobalCorp Data Mesh and governed by Data Product Owners	Mandatory integration with the enterprise data catalog; no shadow data stores
DP-03	Explainability Before Deployment	Any model embedded in an APEX agent must have a documented explainability method (SHAP, LIME, or attention maps) before production approval	Explainability pipeline is a mandatory quality gate in the APEX CI/CD process
DP-04 (AI-DLC)	Prompt Catalogs as Governed Data	Prompt templates, embeddings, and agent feedback logs are first-class data assets governed identically to operational data	ML experiment tracking tool required; prompt versioning in source control; DPO sign-off for sensitive prompts

Technology Principles

ID	Principle Name	Statement	Key Implication
TP-01	Cloud-Native First	Platform components use cloud-native open standards (containers, Kubernetes, OpenAPI, CloudEvents) before proprietary managed services	All components containerised; orchestration via standard Kubernetes or equivalent; no cloud-SDK lock-in
TP-02	Security by Design	Security controls are embedded at design time, never retrofitted after deployment	Security architecture approved before build begins; automated policy-as-code in CI/CD pipeline
TP-03	Observability as First-Class Concern	Every APEX agent emits structured traces, metrics, and logs from day one using OpenTelemetry	OpenTelemetry collector mandatory; centralised observability platform; no agent deployed without a dashboard
TP-04	Cost Transparency	Each AI agent must have a tagged cost centre with monthly spend budget and automated threshold alerts	Resource tagging policy enforced via IaC; FinOps review gate in APEX onboarding checklist
TP-05 (AI-DLC)	Every Engineer Understands Every Line	All AI-generated Infrastructure-as-Code and application code must be reviewed and understood by a human engineer before merge	Mandatory peer review for AI-generated IaC; no automated merge without human approval

PHASE A

Architecture Vision

Define the why and what — secure executive mandate

Phase A creates the compelling case for change, maps the stakeholder landscape, defines architecture scope, and produces the Statement of Architecture Work (SAW) that gives APEX formal mandate. AI-DLC Mob Elaboration sessions can reduce this phase from weeks to days.

Document: Statement of Architecture Work — SAW-APEX-001

DOC REF: SAW-APEX-001 | v1.4 | APPROVED – Group CTO sign-off: 2025-02-14

"By Q4 2026, GlobalCorp will operate a unified AI Agent Platform (APEX) using cloud-native open standards that reduces time-to-market for AI use cases from 9.2 to 2.5 months, achieves full EU AI Act and DORA compliance, and generates \$38 M NPV over five years through platform economics."

Architecture Constraints

Constraint	Source	Architecture Impact
Data must remain within the contractual jurisdiction of the data subject	GDPR / PDPA / LGPD	Multi-region deployment; AI inference endpoints provisioned per jurisdiction
All AI models require Model Risk validation before production	Internal Model Risk Policy (aligned to SR 11-7)	Six-week validation gate per model class; automated evidence generation
Cloud-native open standards preferred over proprietary SDKs	Board Architecture Principle TP-01	All components use container-based deployment; OpenAPI for all agent interfaces
Zero-downtime deployment for critical services	Enterprise SLA Policy	Blue-green deployment pattern mandatory for all Pioneer agents
99.95% platform availability target	Enterprise SLA Policy	Multi-zone deployment + disaster recovery across at least two regions
EU AI Act enforcement from August 2026	European Commission Regulation 2024/1689	High-Risk AI classification and associated controls pre-built into APEX onboarding

Stakeholder Map — Key Players (High Power / High Interest)

Stakeholder	Role	Primary Concerns	APEX Value Proposition	Engagement
Group CTO	Executive Sponsor	AI delivery speed, cost control, regulatory exposure	68% TTM reduction; \$14 M duplicated spend eliminated	Monthly steering committee
VP AI Platform	Programme Owner	Platform adoption, developer experience, roadmap	Self-service marketplace; reusable agent templates	Daily collaboration
Chief Information Security Officer	Security Authority	Data leakage, model poisoning, access control	Centralised guardrails, IAM, end-to-end audit trail	Bi-weekly review
Group Compliance Lead	Regulatory Liaison	EU AI Act, DORA, GDPR obligations	Automated compliance reporting and evidence generation	Monthly evidence pack review
Chief Data Officer	Data Governance Authority	Data mesh integration, lineage, sovereignty	Native data catalog integration; regional data residency enforcement	Bi-weekly architecture review
Industry Regulator	External Oversight	Systemic AI risk, model governance, human oversight	DORA evidence pack; EU AI Act Article 13 transparency controls	Quarterly regulatory briefing

Five Pioneer Domains — Proof of Value

#	Division	Agent Name	Use Case	Regulation	AI-DLC Bolt Estimate
1	Customer Division	RiskScoringAgent	Automated risk-based decisioning with human escalation	Internal Model Risk Policy, GDPR	3 bolts (~6 wks)
2	Enterprise Division	VerificationAgent	Continuous customer verification monitoring and report drafting	AML Directive, FATF 40 Recommendations	3 bolts (~6 wks)
3	Advisory Division	RebalancingAgent	Rule-based portfolio recommendation engine	MiFID II Art.27, GDPR	2 bolts (~4 wks)
4	IT Operations	IncidentAgent	L1/L2 ticket triage, root-cause analysis, auto-remediation	DORA Art.11 ICT Risk	2 bolts (~4 wks)
5	Risk & Compliance	ModelMonitorAgent	Model drift detection, backtesting, regulatory report drafting	EU AI Act Art.9, Internal Policy	3 bolts (~6 wks)

PHASE B

Business Architecture

Capability model, value streams, and organisation design

Phase B produces the Business Architecture for APEX. AI-DLC changes how business capabilities are modelled: capabilities that use AI are no longer static — they 'learn and adapt in motion,' requiring new governance conventions in the Business Capability Map.

Business Capability Map — Gap Summary (BCM-APEX-001)

DOC REF: BCM-APEX-001 | v1.5

L1 Capability Domain	L2 Count	Critical Gaps	APEX Addresses	AI-DLC Adaptive?
1. AI Strategy & Governance	5	4 critical	5 / 5	Partially — policy enforcement via AI agents
2. Agent Development	5	5 critical	5 / 5	YES — AI-DLC Construction phase
3. Agent Orchestration	4	4 critical	4 / 4	YES — AI-DLC Operations phase
4. Knowledge Management	4	4 critical	4 / 4	YES — continuous knowledge base refresh
5. AI Operations (AIOps)	5	5 critical	5 / 5	YES — AI-DLC Operations + Phase H change management
6. Data Integration	4	2 critical	3 / 4	PARTIALLY — Data Mesh remains domain-governed
7. Platform Enablement	4	4 critical	4 / 4	YES — self-service portal scaffolded by AI-DLC

Value Stream: AI Agent Delivery — Baseline vs APEX vs AI-DLC (VSA-APEX-001)

DOC REF: VSA-APEX-001 | v1.1

Stage	Baseline	APEX Target	AI-DLC Duration	AI-DLC Enabler
Intake & Qualification	2–4 weeks	5 business days	1–2 hours	Mob Elaboration; EU AI Act classifier runs automatically at intake
Architecture & Design	4–8 weeks	10 business days	1–3 days (1 bolt)	Mob Construction; AI proposes architecture, humans approve
Build & Integrate	8–16 weeks	3 weeks	1–2 weeks (1–2 bolts)	AI coding assistant generates IaC, serverless functions, agent config
Validate & Govern	6–10 weeks	10 business days	3–5 days	AI-automated RAG evaluation, guardrail tests, compliance evidence generation
Deploy & Monitor	2–4 weeks	1 week	1–2 days	AI-DLC Operations; IaC apply, canary rollout, dashboards auto-provisioned
TOTAL TIME-TO-MARKET	22–42 weeks	~9.5 weeks	~3–4 weeks	AI-DLC compounds productivity gains across every stage

Lean Value Waste Elimination Analysis

Waste Type	Current Annual Cost	APEX Reduction	AI-DLC Further Reduction	Residual
Rework — duplicate agent patterns built per team	\$3.2 M	–80% via platform reuse	–15% via AI pattern suggestion	\$0.2 M
Waiting — manual compliance review queue (avg. 6 weeks)	\$4.8 M	–60% via automated pipeline	–30% via AI-generated evidence packs	\$0.5 M
Overprocessing — each team builds own observability stack	\$2.1 M	–90% via shared observability	–5% via AI-generated dashboards from schema	\$0.1 M

Waste Type	Current Annual Cost	APEX Reduction	AI-DLC Further Reduction	Residual
Inventory — 14 abandoned partially-built agent projects	\$1.9 M	–70% via marketplace discovery	–20% via AI-DLC fast bolt completion	\$0.3 M
Defects — untested agents causing production incidents	\$2.0 M	–75% via mandatory CI/CD gate	–20% via AI-generated comprehensive test suites	\$0.1 M
TOTAL	\$14.0 M/yr	\$7.6 M reduction	\$1.5 M additional reduction	\$1.1 M residual

PHASE C

Information Systems Architecture

Data architecture and cloud-native application component design

Phase C introduces AI-native data entities — embeddings, vector stores, prompt catalogs, feedback logs — that do not exist in traditional TOGAF data models, and agent-oriented application patterns that require new architectural conventions. All designs use cloud-native open standards.

Data Classification Framework (DAD-APEX-001)

DOC REF: DAD-APEX-001 | v1.3

Class	Description	Examples	Guardrail Action	Retention
C4 — Highly Restricted	PII + highly sensitive data	Government IDs, account numbers, biometric data	BLOCK or ANONYMIZE in all agent logs and traces	7 years (DORA mandate)
C3 — Confidential	Sensitive internal business data	Proprietary models, strategic parameters, risk thresholds	Redact in all observability traces and log exports	5 years
C2 — Internal	Non-sensitive operational data	Agent performance metrics, usage statistics, cost logs	Standard structured logging	3 years
C1 — Public	Regulatory and market reference data	EU AI Act text, public market rates, open standards	No restriction required	Indefinite
C-AI — AI Assets (AI-DLC NEW)	Prompt catalogs, embeddings, feedback logs, model lineage	Versioned prompt templates, vector embeddings, RLHF datasets	Inherit from source data class; version-controlled; DPO sign-off for C3/C4-adjacent prompts	Per source data class rules

AI-Native Data Entities (AI-DLC Extension to TOGAF Data Architecture)

These entities must be treated as first-class governed artefacts — not infrastructure configuration:

Entity	Description	Governance Owner	Open-Standard Tooling	Risk if Ungoverned
Prompt Catalog	Versioned prompt templates per agent; treated as regulated code and data	Platform AI Engineer + DPO	ML experiment tracking (MLflow Prompt Registry or equivalent) + source control	Prompt drift changes agent behaviour without visibility; PII leakage through unreviewed prompts
Embedding Store	Vector representations of knowledge base content; versioned per refresh cycle	Data Mesh Domain Owner	Open-source vector database (e.g. Weaviate, Qdrant, OpenSearch)	Stale embeddings cause hallucinations; version mismatch breaks retrieval quality
Feedback Log	Human-in-Loop decisions, corrections, and escalation reasons from all agent interactions	Model Risk + Compliance	Immutable object storage with WORM policy + data catalog metadata	Model drift goes undetected; EU AI Act Art.9 evidence gaps; regulatory audit failures
Agent Memory	Cross-session context store; session summaries; entity memory for long-running workflows	Platform Engineering + DPO	Key-value store with TTL policy (e.g. Redis, DynamoDB equivalent)	PII retention violation if session memory TTL is not enforced; GDPR Art.5 breach risk
Model Lineage Record	Full chain from training data through model version to production deployment record	Model Risk Management	ML model registry (MLflow Model Registry or equivalent)	EU AI Act Art.13 transparency breach; unable to explain agent decisions to regulators
RAG Retrieval Log	Which knowledge base chunks were retrieved for each agent decision, with similarity scores	Compliance + Internal Audit	Structured log store with query capability; retained in data catalog	Cannot explain or audit agent decision basis; BCBS 239 data lineage gap

Knowledge Base Architecture — Cloud-Native RAG Design

KB ID	Purpose	Sources	Chunking Strategy	Embedding Model	Refresh Cadence
KB-001	Policy & Compliance	Regulatory handbooks, internal policy manuals, sector-specific guidelines	Recursive, 512 tokens, 10% overlap	Open-source sentence-transformer or commercial embedding API (provider-agnostic)	Nightly for internal; weekly for regulatory sources
KB-002	Customer Verification Rules	AML directive guidance, FATF 40 Recommendations, internal verification procedures	Semantic, 1024 tokens, 15% overlap	Same embedding model as KB-001 for consistency	Weekly regulatory source monitoring; on-change trigger for internal
KB-003	Product & Suitability	Product catalogue, suitability rules, risk classification definitions	Recursive, 512 tokens, 10% overlap	Domain-fine-tuned embedding model if available	Real-time trigger on product data change events
KB-004	IT Operations Runbooks	Incident runbooks, DORA incident response playbooks, on-call guides, troubleshooting trees	Semantic, 768 tokens, 12% overlap	General-purpose embedding model	On-change trigger; validated by SRE team before publish

Application Architecture — Cloud-Native Component Model (AAD-APEX-001)

DOC REF: AAD-APEX-001 | v1.4

Component	Cloud-Native Technology	Design Pattern	SLA Target	AI-DLC Note
APEX Developer Portal	SPA framework (React/Vue) + API Gateway (any cloud or Kong/Nginx)	Backend-for-Front end (BFF)	99.9% / p99 < 500 ms	UI scaffolded by AI coding assistant in Mob Construction in 1 bolt
APEX Control Plane	Python FastAPI in OCI containers; orchestrated by Kubernetes	Microservices (5 bounded contexts)	99.9% / p99 < 500 ms	FastAPI skeleton and unit tests AI-generated in bolt 1; peer reviewed
AI Agent Orchestration Engine	Provider-agnostic agentic framework (LangGraph, AutoGen, or cloud-managed equivalent)	Supervisor-Worker multi-agent pattern	99.95% / p99 < 3 s	Agent YAML and prompt templates generated via AI-DLC; ARB reviews pattern
Serverless Tool Functions	OCI function runtime (container-based serverless: OpenFaaS, Knative, or cloud-native FaaS)	Action Group / Tool Sidecar pattern	99.95% / p99 < 1 s	Each tool function AI-generated; mandatory human review of every function before merge
Human Review Portal	Custom React UI + workflow engine (Temporal, Camunda, or managed equivalent)	Human-in-the-Loop (HITL) review workflow	99.5% / SLA defined per agent risk tier	Workflow config AI-generated; custom review UI built in mob session
Observability Stack	OpenTelemetry Collector → Prometheus + Grafana or any OTEL-compatible backend	Centralised observability; push-based metrics and traces	99.5% ingestion / real-time dashboards	Dashboard-as-code generated by AI-DLC from agent schema definition
Compliance Reporter	Python data pipeline + open BI tool (Apache Superset, Metabase, or equivalent)	Batch report generation; event-driven for DORA incident triggers	99.5% / daily scheduled run	EU AI Act and DORA report templates AI-generated; Compliance team reviews

API Catalogue

API Name	Version	Style	Auth Method	SLA	Owner
Agent Registry API	v2	REST / OpenAPI 3.1	OAuth 2.0 (APEX scope)	99.9% / p99 < 200 ms	Platform Eng
Agent Invocation API	v1	REST + SSE streaming	IAM / mTLS	99.95% / p99 < 3 s	Platform Eng
Knowledge Base API	v1	REST / JSON	IAM / mTLS	99.9% / p99 < 500 ms	AI/ML Eng
Onboarding API	v1	REST / JSON	OAuth 2.0	99.5% / p99 < 1 s	Platform Eng
Chargeback API	v1	REST / JSON	mTLS	99.9% monthly	FinOps
Compliance Report API	v1	REST / JSON	OAuth 2.0 + MFA	99.5% daily run	Compliance

PHASE D

Technology Architecture

Cloud-native infrastructure, security, CI/CD, and observability

Phase D specifies the technology architecture using cloud-native, vendor-agnostic patterns. The architecture is designed to be deployable on any major cloud provider or on-premises Kubernetes cluster without architectural rework. Cloud-specific managed services are treated as implementation choices made during Phase F, not Phase D.

Multi-Region Deployment Model (TAD-APEX-001)

DOC REF: TAD-APEX-001 | v2.0

Deployment Region	Role	Data Types Processed	Key Cloud-Native Services	Regulatory Basis
Region 1 (Primary — EU/UK)	PRIMARY active	EU/UK personal data, sensitive records	Kubernetes cluster; managed vector store; object storage (WORM policy); managed relational DB; event streaming	GDPR, UK GDPR, DORA
Region 2 (DR — EU)	Disaster Recovery / failover	DR replica of Region 1 data only; no new processing	Kubernetes cluster (standby); object storage (CRR target); managed DB replica	DORA Art.11 ICT resilience; GDPR Art.32
Region 3 (Americas)	AMERICAS active	US/LATAM institutional and customer data	Mirror of Region 1 topology; independent control plane	CCPA, LGPD, FINRA (if applicable)
Region 4 (APAC)	APAC active	Singapore, HK, Australia customer data	Mirror of Region 1 topology; local vector store; local object storage	PDPA (SG), PIPL (CN), Privacy Act (AU)

Cloud-Native AI Agent Orchestration Configuration

Parameter	Value / Setting	Rationale
Foundation Model	Vendor-agnostic: selected per task type (reasoning, summarisation, classification) with benchmark evaluation	No single-vendor lock-in; model selection updated quarterly based on benchmarks on GlobalCorp eval suite
Session Timeout	1800 seconds (30 minutes)	Balances UX continuity with PII data minimisation obligation under GDPR Art.5
Memory Architecture	Session summary memory enabled; entity memory for long-running workflows	Supports multi-step business processes without overloading model context window
Content Guardrails	PII detection (ANONYMIZE), sensitive topic filtering (BLOCK), grounding threshold 0.75	Guardrails configured in open-source safety layer (NeMo Guardrails or equivalent); not vendor-specific
Grounding Threshold	0.75 minimum — reject ungrounded responses	Prevents hallucinations in regulated decisions; RAG retrieval log written for every response
Agent Invocation Path	Private network only — no public internet egress from agent runtime	Security architecture principle TP-02; enforced via Kubernetes NetworkPolicy
Distributed Tracing	OpenTelemetry traces on every agent interaction; 100% sampling for regulated agents	DORA Art.11 audit trail; EU AI Act Art.13 traceability; basis for BCBS 239 data lineage

Security Architecture Summary

Security Domain	Control	Cloud-Native Implementation	Compliance Basis
Identity & Access	Zero-trust IAM; short-lived credentials for all service accounts	OIDC-based identity federation; service mesh mTLS (Istio or Linkerd); RBAC via Kubernetes	DORA Art.9, ISO 27001, NIST SP 800-207
Encryption at Rest	AES-256 encryption with customer-managed keys; separate key per data class	KMS abstraction layer (cloud-provider KMS or HashiCorp Vault); key rotation every 365 days	GDPR Art.32, PCI DSS 3.2.1

Security Domain	Control	Cloud-Native Implementation	Compliance Basis
Encryption in Transit	TLS 1.3 minimum enforced across all internal and external communication paths	Istio/Linkerd service mesh enforces mTLS internally; Ingress controller enforces TLS 1.3 externally	FIPS 140-2, DORA, ISO 27001
Network Isolation	Zero-trust network policy; agent runtime has no public internet egress	Kubernetes NetworkPolicy; private endpoints for all external API calls; egress via controlled proxy	TP-01, Security Policy, DORA
Threat Detection	Runtime anomaly detection; container image scanning; SAST/DAST in CI/CD	Falco (runtime security); Trivy or Grype (image scanning); OWASP ZAP in pipeline	DORA Art.11, SOC 2 Type II
Agent Safety	Content guardrail layer; PII anonymisation; grounding enforcement	NeMo Guardrails or equivalent open framework; integrated into agent invocation path	EU AI Act Art.9, BP-02, DP-01
Prompt Injection Defence	Input sanitisation and validation on all agent inputs before model invocation	Validation middleware in agent gateway; OWASP LLM Top 10 controls; red-team testing quarterly	OWASP LLM Top 10, Security Policy
Secrets Management	All secrets stored in a secrets manager; zero hardcoded credentials anywhere	HashiCorp Vault or cloud-native secrets manager; enforced by Checkov in CI/CD pipeline	CIS Kubernetes Benchmark, DORA

CI/CD Pipeline — AI-DLC Enhanced (Cloud-Native)

Stage	Open-Source / Cloud-Native Tools	AI-DLC Enhancement	Gate Type
1. Code Generation	AI coding assistant (provider-agnostic); IDE plugin; source control (Git)	AI generates IaC, serverless functions, agent config from semantic context	Human review mandatory per TP-05
2. Static Analysis	Checkov (IaC), Semgrep (SAST), yamllint, OWASP Dependency Check	AI explains failures and proposes remediation within the coding assistant	Automated block on any HIGH or CRITICAL finding
3. Prompt Regression	RAG evaluation framework (RAGAS or equivalent); custom eval suite	AI-generated test cases run against baseline; flagged regressions block merge	Automated; >90% score required to pass
4. Guardrail Testing	NeMo Guardrails test harness; adversarial prompt library	AI generates adversarial prompts to probe guardrail coverage and edge cases	Automated; 100% pass rate required
5. Model Risk Gate	Ticket-based approval workflow (Jira, Linear, or equivalent)	AI generates model risk evidence pack draft; compliance team reviews and approves	Manual human sign-off; blocks deployment until approved
6. Non-Prod Deploy	Helm + Argo CD (GitOps); Kubernetes; environment-parity staging	AI-DLC Operations: manifests applied automatically; AI-generated smoke tests run	Automated deployment + architect review of results
7. Canary Deploy	Argo Rollouts or Flagger (progressive delivery); Prometheus metrics	AI monitors canary SLOs; triggers automatic rollback if p99 latency or error rate breaches threshold	Automated with human override capability
8. Post-Deploy	OpenTelemetry + Grafana; cost tagging validator; compliance record writer	AI generates stakeholder summary; cost tags verified; DORA evidence record auto-created	Automated compliance record; FinOps notified of spend delta

PHASES E-H

Opportunities, Migration, Governance & Change

Roadmap, delivery planning, compliance contracts, and change management

Phase E — Architecture Roadmap (ROAD-APEX-001)

DOC REF: ROAD-APEX-001 | v2.1

AI-DLC compresses the roadmap significantly. Traditional work packages measured in months are replaced by bolts (hours to days). The table maps each work package to its AI-DLC bolt estimate and delivery horizon:

Work Package	WP-ID	AI-DLC Bolts	Calendar Weeks	Horizon
Cloud-native multi-region infrastructure (Kubernetes, networking, IAM, KMS)	WP-001	2 bolts	4 weeks	H1
AI Agent Orchestration Platform setup (framework, guardrails, KB-001)	WP-002	2 bolts	4 weeks	H1
CI/CD pipeline with AI-DLC tooling integration	WP-004	1 bolt	2 weeks	H1
Security baseline (all controls — zero-trust, mTLS, secrets, scanning)	WP-008	2 bolts	4 weeks	H1
Observability stack (OpenTelemetry, Prometheus, Grafana, alerting)	WP-007	1 bolt	2 weeks	H1
Pioneer 1 — Customer Risk Scoring Agent	WP-005	3 bolts	8 weeks	H1
Pioneer 2 — Customer Verification Agent	WP-006	3 bolts	8 weeks	H1
Pioneer 3 — Advisory Rebalancing Agent	WP-009	2 bolts	6 weeks	H2
Pioneer 4 — IT Incident Response Agent	WP-010	2 bolts	5 weeks	H2
Pioneer 5 — Model Monitoring & Compliance Agent	WP-011	3 bolts	6 weeks	H2
Self-service onboarding portal and agent marketplace	WP-012	3 bolts	8 weeks	H2
EU AI Act compliance pack (automated classification, evidence, reporting)	WP-015	2 bolts	5 weeks	H2

Phase E — Solution Building Blocks Catalogue (SBB-APEX-001)

DOC REF: SBB-APEX-001 | v1.2

SBB	Decision	Cloud-Native Solution	AI-DLC Role
Agent Orchestration Engine	BUY or OPEN-SOURCE	LangGraph, AutoGen, CrewAI, or cloud-managed agentic service (provider-evaluated in Phase F)	AI-DLC generates agent YAML, tool definitions, and action group config
Foundation Models	BUY (API-accessed)	Evaluated per task: open-weight (Llama, Mistral) or commercial API; no single-vendor lock-in	AI-DLC Construction: AI selects model per task type from pre-approved catalogue
Vector Store	OPEN-SOURCE or MANAGED	Weaviate, Qdrant, Milvus, or OpenSearch (provider-evaluated in Phase F)	AI-DLC generates index schema, query patterns, and refresh pipeline
Agent Control Plane	BUILD	Python FastAPI on Kubernetes (OCI containers; Helm chart)	AI-DLC Construction builds entire service in 2 bolts; peer reviewed
Developer Portal	BUILD	React SPA + API Gateway (Kong, Nginx, or cloud-native ingress)	AI-DLC scaffolds UI components in Mob Construction in 1 bolt

SBB	Decision	Cloud-Native Solution	AI-DLC Role
Human Review Portal	BUILD on open framework	Custom React UI + Temporal or Camunda workflow engine	AI-DLC generates workflow definitions; custom UI mob-built in 1 bolt
Observability Platform	OPEN-SOURCE	OpenTelemetry Collector + Prometheus + Grafana + Loki (full LGTM stack)	AI-DLC generates dashboards-as-code (Grafana JSON) from agent metric schema
CI/CD Platform	REUSE	GitLab CI or GitHub Actions + Argo CD (GitOps); existing enterprise tooling	AI-DLC extends existing pipeline with agent-specific quality gates
Safety / Guardrails Layer	OPEN-SOURCE	NVIDIA NeMo Guardrails or Guardrails AI; configured per agent risk tier	AI-DLC generates guardrail rule sets; red-team tests AI-generated
Compliance Reporter	BUILD	Python data pipeline + Apache Superset or Metabase for reporting UI	AI-DLC generates EU AI Act and DORA report templates from schema

Phase F — Migration Strategy: Existing Siloed AI Tools (IMP-APEX-001)

DOC REF: IMP-APEX-001 | v1.5

Category	Count	Action	AI-DLC Migration Path	Target
ADOPT — Cloud-native; add APEX governance wrapper	8 tools	Wrap in APEX tool interface; add guardrail and observability layers	AI-DLC Operations: IaC wrapping and registration in 1 bolt per tool	Q3 2025
ADAPT — Works well but needs APEX integration layer	6 tools	Expose as APEX tool via standardised OpenAPI interface; APEX orchestrates	AI-DLC Construction builds adapter function in Mob Construction session	Q4 2025
REPLACE — Capability fully duplicated by APEX Pioneer agent	7 tools	Pioneer agent replaces in parallel-run; legacy decommissioned after 4-week parallel	Parallel-run then canary cutover; AI monitors SLOs and triggers auto-rollback if needed	Q1 2026
RETIRE — Unused or insufficient value to migrate	2 tools	Immediate decommission; final audit log snapshot archived immutably	AI-DLC Operations: IaC teardown + automated evidence generation for audit trail	Q2 2025

Phase G — Architecture Governance: Tiered Conformance Model (CAF-APEX-001)

DOC REF: CAF-APEX-001 | v1.1 – AI-DLC Impact: Continuous Automated Conformance

Tier	Checks Included	Method	Frequency	Human Involvement
Tier 1 — Automated (always-on)	IaC policy checks (Checkov), container scanning (Trivy), resource tag compliance, RAG evaluation score, cost budget alerts	CI/CD pipeline + Kubernetes admission controllers + scheduled validation jobs	Every commit and every 6 hours	Alert-only; human reviews consolidated findings dashboard daily
Tier 2 — Architect-in-the-Loop	Architectural conformance (design vs blueprint), new agent pattern review, data classification audit	Automated check triggers Slack or Teams notification to architect on-call with finding summary	Every deployment + daily digest	Architect reviews and approves within 4 business hours
Tier 3 — ARB Review	Significant deviations from approved architecture, new solution building block proposals, architectural debt threshold breach	Formal ARB session — now weekly (accelerated from bi-weekly due to AI-DLC velocity)	Weekly; escalated immediately for critical findings	Full ARB; human decision authority retained in all cases
Tier 4 — Steering Committee	Strategic deviations, budget impacts above approved threshold, regulatory findings, programme risk escalations	Formal steering pack — AI-generated draft reviewed by EA lead before presentation	Monthly; emergency session within 48 hours for DORA-reportable incidents	Executive committee; final authority

Phase H — Change Management: Continuous Change Intelligence (ACHG-APEX-001)

DOC REF: ACHG-APEX-001 | v1.2 – AI-DLC Impact: Automated Landscape Monitoring

Change Type	AI-DLC Monitoring Method	Response SLA	Realistic Example
Type 1 — Technology-driven (new model release, new framework version)	AI agent monitors foundation model release channels and open-source project feeds; flags architecture impact automatically	Architecture impact assessment within 5 business days of detection	New open-weight model (e.g. Llama 4) outperforms current model on GlobalCorp eval suite by 18%; ARB approves model swap within 3 days
Type 2 — Regulatory-driven (new regulation, enforcement milestone)	Regulatory feed monitored daily by compliance-monitoring agent; impact classified against APEX controls automatically	Compliance impact assessment within 48 hours; control gap remediation plan within 5 business days	EU AI Act Art.6 enforcement date reached; APEX onboarding flow updated to enforce High-Risk classification automatically
Type 3 — Business-driven (new domain team requests platform access)	Domain team submits agent onboarding request via self-service portal; standard intake workflow triggered	Intake decision within 5 business days; architecture review within 10 business days	New Procurement Division requests APEX onboarding for contract analysis agent
Type 4 — Simplification (underutilised agent or deprecated pattern)	APEX usage analytics identifies agents below utilisation threshold quarterly; flags for decommission review	Decommission plan within 30 days of flag; executed within 60 days with stakeholder sign-off	Pioneer 1 agent v1.2 superseded by v2.0; v1.2 decommissioned after parallel-run validation

REQ MGMT

Requirements Management (Continuous)

Living requirements — the heartbeat of the APEX ADM

Requirements Management is the continuous central process of the TOGAF ADM wheel. AI-DLC's Semantic Context Building and persistent context storage transform this from a periodic document update into a living, AI-maintained traceability system. GlobalCorp maintains 147 architecture requirements in the ARS; a representative sample follows.

Architecture Requirements Specification — ARS-APEX-001 (Sample)

DOC REF: ARS-APEX-001 | v2.3 | Living Document – 147 total requirements

Req ID	Category	Statement	Priority	Source	Status
BR-001	Business	Platform supports at minimum 120 concurrent agent interactions per region under normal load	Must	Capacity model	In Design
BR-003	Business	All agent decisions logged with full context and queryable for a minimum of 7 years	Must	DORA Art.11	In Design
BR-006	Business	EU AI Act risk tier automatically classified for every agent at the point of intake into the onboarding workflow	Must	EU AI Act Art.6	In Design
DR-001	Data	Personal data never persisted in agent conversation logs, session memory, or observability traces	Must	GDPR Art.5	In Design
DR-AI-01	Data (AI-DLC)	Prompt templates versioned in source control; any change triggers automated regression test suite before merge	Must	TP-05 / AI-DLC	Backlog
DR-AI-02	Data (AI-DLC)	Embeddings versioned per knowledge base refresh cycle; superseded versions retained for 90 days minimum to support rollback	Must	DR-003 / EU AI Act	Backlog
AR-001	Application	Agent Control Plane API achieves p99 response time below 500 ms under a load of 500 requests per second	Must	SLA Policy	In Design
TR-001	Technology	All AI model invocations travel exclusively over private network paths; no public internet egress from agent runtime pods	Must	Security Policy / TP-02	In Design
TR-AI-01	Technology (AI-DLC)	All AI-generated IaC and application code reviewed and approved by a human engineer before merge; automated merge blocked	Must	TP-05 / AI-DLC	In Design
SR-001	Security	Customer-managed encryption keys rotated on a maximum 365-day cycle; rotation enforced by automated compliance check	Must	GDPR Art.32 / ISO 27001	In Design
SR-003	Security	Red-team adversarial testing of each agent completed and documented before the agent is approved for production deployment	Must	AI Risk Policy / OWASP LLM	In Review
CR-001	Compliance	EU AI Act risk classification documented and stored in the agent registry for every agent across all environments	Must	EU AI Act Art.9	In Design
CR-002	Compliance	Model explainability method (SHAP, LIME, or equivalent) documented and evidence artefact generated before production approval	Must	EU AI Act Art.13	In Design
CR-003	Compliance	DORA 72-hour ICT major incident notification workflow automated and tested quarterly under simulated conditions	Must	DORA Art.17	In Review
CR-AI-01	Compliance (AI-DLC)	All architecture artefacts generated with AI assistance are tagged with AI-DLC metadata (model version, prompt hash) for audit trail	Should	EA Governance / EU AI Act	Backlog

APPENDIX A

Regulatory Cross-Reference Matrix

APEX compliance posture across all applicable regulatory frameworks

APEX is designed to be compliant-by-default across all applicable regulatory frameworks. The matrix below maps each regulation to the specific architecture controls that satisfy its requirements and the evidence artefacts that demonstrate compliance:

Regulation	Key Articles / Obligations	APEX Architecture Control	Evidence Artefact
EU AI Act (2024/1689)	Art.6 (High-Risk classification), Art.9 (Risk Mgmt System), Art.13 (Transparency), Art.17 (Quality Mgmt), Art.72 (Penalties up to €35 M)	EU AI Act risk classifier at intake (WP-015); model explainability pipeline in CI/CD; full agent decision audit trail; AI-DLC artefact metadata tagging	EU AI Act Compliance Pack (auto-generated by Pioneer 5 agent); Model Risk Opinions; Audit trail exports
DORA (EU) 2025	Art.9 (ICT security policy), Art.11 (ICT risk management), Art.17 (major incident notification within 72 hrs), Art.26 (Threat-Led Penetration Testing)	Zero-trust security architecture; immutable WORM audit logs; automated 72-hour incident workflow via IT Incident Agent (Pioneer 4); quarterly penetration testing	DORA Evidence Pack (quarterly); Security scan reports; Incident runbook tests; TLPT results
GDPR / UK GDPR	Art.5 (data minimisation), Art.25 (privacy by design), Art.32 (appropriate security), Art.46 (third-country transfers prohibited without safeguards)	PII ANONYMIZE in all guardrail layers; data residency enforced per region; AES-256 CMK encryption; no cross-border data transfer without adequacy decision	Article 30 records of processing; Data Flow Maps per Pioneer agent; Guardrail configuration evidence
Internal Model Risk Policy (aligned SR 11-7)	Model validation before production, documentation of assumptions and limitations, ongoing performance monitoring, independent review	Model Risk opinion gate in CI/CD pipeline (WP-008); SageMaker-equivalent explainability; Model Monitoring Agent (Pioneer 5) for continuous drift detection	Model Risk Opinions per Pioneer agent; Explainability reports; Drift monitoring dashboards
BCBS 239	Data lineage for risk aggregation, accuracy, completeness, and timeliness of risk data reporting	Data catalog lineage integration; RAG retrieval logs trace every agent decision to source data; agent decision audit trail queryable via BI tool	Data catalog lineage reports; RAG retrieval log exports; Agent decision audit trail samples
MiFID II Art.27	Best execution obligations; suitability assessment for investment advice; record-keeping	Human-in-the-Loop mandatory for all investment advisory outputs; suitability rules in KB-003; full session and decision logging	Human review audit logs (HITL portal); Suitability assessment records; Session logs per advisory interaction

APPENDIX B

Cloud-Native Service Mapping

Vendor-agnostic architecture components mapped to cloud-provider equivalents

The APEX architecture is designed against cloud-native patterns and open standards. The table below maps each architectural capability to its open-source/open-standard reference implementation and example cloud-provider managed equivalents. Final provider selection occurs in Phase F.

APEX Capability	Open-Standard / OSS Reference	Cloud Provider A Equivalent	Cloud Provider B Equivalent	Cloud Provider C Equivalent
Container Orchestration	Kubernetes (CNCF)	EKS (AWS)	GKE (GCP)	AKS (Azure)
Serverless Functions	Knative / OpenFaaS	AWS Lambda	Cloud Functions (GCP)	Azure Functions
AI Agent Orchestration	LangGraph / AutoGen / CrewAI	Bedrock Agents (AWS)	Vertex AI Agents (GCP)	Azure AI Foundry Agents
Foundation Model API	Ollama (local) / open-weight via API	Bedrock (Claude, Titan)	Vertex AI (Gemini, Claude)	Azure OpenAI (GPT-4o, o1)
Vector Store	Weaviate / Qdrant / Milvus	OpenSearch Serverless (AWS)	AlloyDB + pgvector (GCP)	Azure AI Search
Object Storage	MinIO / Ceph (S3-compatible API)	S3 (AWS)	Cloud Storage (GCP)	Azure Blob Storage
Managed Relational DB	PostgreSQL (self-managed on Kubernetes)	Aurora PostgreSQL (AWS)	Cloud SQL PostgreSQL (GCP)	Azure Database for PostgreSQL
Event Streaming	Apache Kafka / Redpanda	Kinesis / MSK (AWS)	Pub/Sub / Dataflow (GCP)	Azure Event Hubs / Service Bus
Workflow Orchestration	Temporal / Apache Airflow / Argo Workflows	Step Functions (AWS)	Cloud Composer / Workflows (GCP)	Azure Logic Apps / Durable Functions
Key Management	HashiCorp Vault	KMS (AWS)	Cloud KMS (GCP)	Azure Key Vault
Observability — Metrics	Prometheus	CloudWatch Metrics (AWS)	Cloud Monitoring (GCP)	Azure Monitor
Observability — Tracing	Jaeger / Tempo (OpenTelemetry backend)	X-Ray (AWS)	Cloud Trace (GCP)	Azure Application Insights
Observability — Logging	Loki / Elasticsearch / OpenSearch	CloudWatch Logs (AWS)	Cloud Logging (GCP)	Azure Monitor Logs
Observability — Dashboards	Grafana	CloudWatch Dashboards (AWS)	Cloud Monitoring Dashboards	Azure Dashboards
GitOps / CD	Argo CD / Flux	CodePipeline (AWS)	Cloud Deploy (GCP)	Azure DevOps Pipelines
Service Mesh / mTLS	Istio / Linkerd (CNCF)	App Mesh (AWS)	Traffic Director (GCP)	Azure Service Mesh (Istio)
ML Experiment Tracking	MLflow	SageMaker Experiments (AWS)	Vertex AI Experiments (GCP)	Azure ML Experiments
AI Safety / Guardrails	NVIDIA NeMo Guardrails / Guardrails AI	Bedrock Guardrails (AWS)	Vertex AI safety filters (GCP)	Azure AI Content Safety
Infrastructure as Code	Terraform / OpenTofu (open-source fork)	CDK (AWS)	Deployment Manager / Terraform	Bicep / ARM Templates

APPENDIX C

Glossary

Key terms — TOGAF 10, AI-DLC, cloud-native, and regulatory

ADM	Architecture Development Method — the iterative process at the core of TOGAF 10; all phases may run concurrently or be scoped independently.
Agent Memory	Persistent context stored across an AI agent's conversation sessions; must be governed with TTL policies to comply with data minimisation principles.
AI-DLC	AI-Driven Development Lifecycle — AWS methodology (July 2025) that positions AI as a central collaborator throughout all phases of software development.
APEX	AI Platform of Platforms — the GlobalCorp programme to build a shared, governed, cloud-native AI agent infrastructure.
ARB	Architecture Review Board — governance body that approves architecture decisions, manages conformance, and maintains the architecture principles.
Bolt	AI-DLC unit of development intensity; equivalent to an Agile sprint but measured in hours to days, not weeks. Replaces the sprint in AI-DLC vocabulary.
CNCF	Cloud Native Computing Foundation — open-source foundation hosting Kubernetes, Prometheus, OpenTelemetry, Argo CD, and other key cloud-native projects.
DORA	EU Digital Operational Resilience Act — regulation requiring ICT risk management, major incident notification within 72 hours, and Threat-Led Penetration Testing.
Embedding	A fixed-size vector representation of text or data, used in RAG systems for semantic similarity search; a first-class data entity in AI-DLC architecture.
EU AI Act	European AI Regulation 2024/1689 — classifies AI systems by risk level and mandates transparency, human oversight, explainability, and quality management for High-Risk AI.
GitOps	Operational model where the desired state of infrastructure and applications is declared in a Git repository and continuously reconciled by an automated operator (e.g. Argo CD).
Guardrails	A safety layer that intercepts agent inputs and outputs to enforce content policies, anonymise PII, filter harmful content, and verify grounding of responses.
Human-in-the-Loop	Architecture pattern where AI agent outputs above a defined risk threshold are routed to a human reviewer before being acted upon or returned to the end user.
IaC	Infrastructure as Code — the practice of defining all infrastructure in version-controlled declarative files (Terraform, Helm, Kustomize); enforced by TP-05.
Knowledge Base	A curated, chunked, and embedded collection of documents that agents retrieve context from at inference time (Retrieval-Augmented Generation).
Mob Construction	AI-DLC ritual where the entire cross-functional team works in a shared session with AI to build solutions — architecture, code, and tests — in real time.
Mob Elaboration	AI-DLC ritual where the team works with AI to transform business intent into detailed requirements, stories, and architecture units; replaces separate planning sessions.
OTEL / OpenTelemetry	Open-source observability framework providing vendor-neutral APIs and SDKs for traces, metrics, and logs; the mandatory observability standard for all APEX agents.
Pioneer Domain	One of five GlobalCorp divisions selected as proof-of-value implementations for the APEX platform; each delivers one production AI agent.
Prompt Catalog	A versioned, governed library of prompt templates used by APEX agents; treated identically to application code — version-controlled, tested, and peer-reviewed.
RAG	Retrieval-Augmented Generation — technique where an AI agent retrieves relevant context from a knowledge base before generating a response, reducing hallucination.
RAGAS	RAG Assessment framework — evaluates RAG pipeline quality across faithfulness, answer relevance, context recall, and context precision metrics.
SAW	Statement of Architecture Work — TOGAF document that formally initiates an architecture engagement, defining scope, constraints, resources, and executive sponsor.
SBB	Solution Building Block — a specific implementation of an Architecture Building Block, defining what will be built, bought, or reused for a given capability.
Semantic Context	AI-DLC concept: rich, structured context provided to AI agents through steering files so they apply organisation-specific standards automatically and consistently.

Unit of Work	AI-DLC equivalent of an Agile Epic; a significant deliverable decomposed into bolts; typically measured in days to weeks in duration.
Zero-Trust	Security model where no network location or identity is inherently trusted; every request is authenticated, authorised, and validated regardless of origin.

Document Control — Version History

Version	Date	Author	Change Summary	Approver
1.0	2025-01-15	Enterprise Architect	Initial draft — Phases A through D	ARB
2.0	2025-03-01	Enterprise Architect	Full ADM coverage added; Architecture Contracts and Governance sections	Group CTO
3.0	2026-04-01	Enterprise Architect	Cloud-native edition: AWS dependencies removed; AI-DLC impact analysis integrated; table wrapping fixed	Group CTO

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